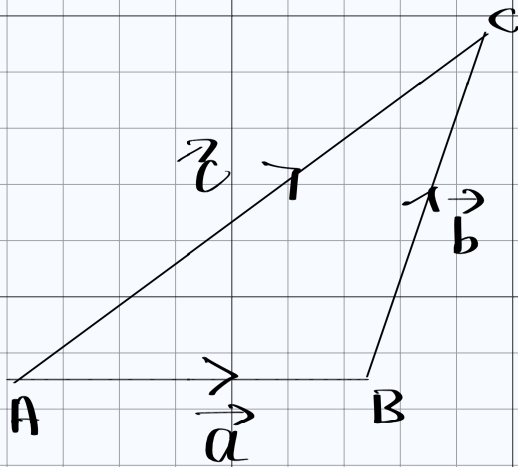


Triangle Law of Vector Addition-



Let say I am basically moving from A to B and then B to C

the resultant or the movement from A to B and then B to C will be equal from movement A to C directly.

That is called triangle law of vector addition.

$$\vec{AB} + \vec{BC} = \vec{AC}$$

In other words two sides of

Side taken in and
order is equal to third side
of triangle taken from opposite
order.

$$\vec{AB} + \vec{BC} + \vec{CA} = 0$$

$$\vec{AB} + \vec{BC} = -\vec{CA}$$

$$\boxed{\vec{AB} + \vec{BC} = \vec{AC}}$$

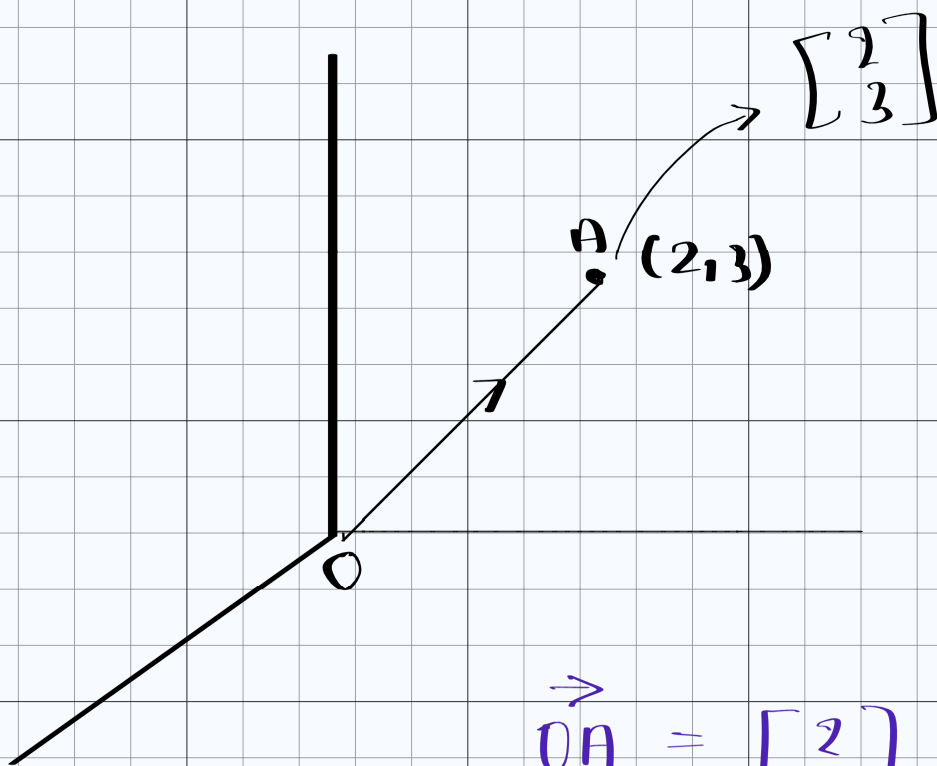
Before going to learn where
we going to use triangle
law of vector addition let's
see something else regarding
vector -

Let say there is a point
in space $A(2, 3)$ we can write
it as vector notation $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$

But there is also one thing

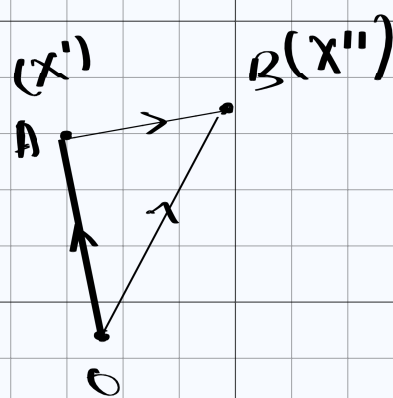
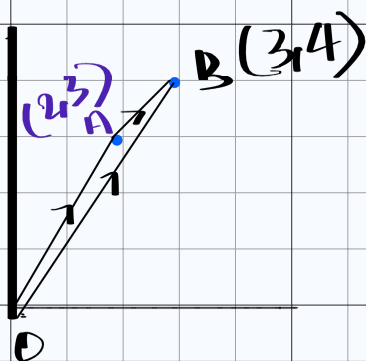
A point is represented by

a vector from origin to that point (position vector)



$$\vec{OA} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} = 2\hat{i} + 3\hat{j}$$

Now let say you are given two points A (2, 3) & B (3, 4) then what is $\vec{AB}$?



$$OB + BA + AO = 0$$

$$OB + AO = -BA$$

$$\vec{OB} + \vec{AO} = \vec{AB}$$

OR

$$\vec{AB} = \vec{OB} - \vec{OA}$$

$$= \vec{x''} - \vec{x'}$$

$$\vec{AB} = \vec{OB} - \vec{OA}$$

$$= \begin{bmatrix} 3 \\ 4 \end{bmatrix} - \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

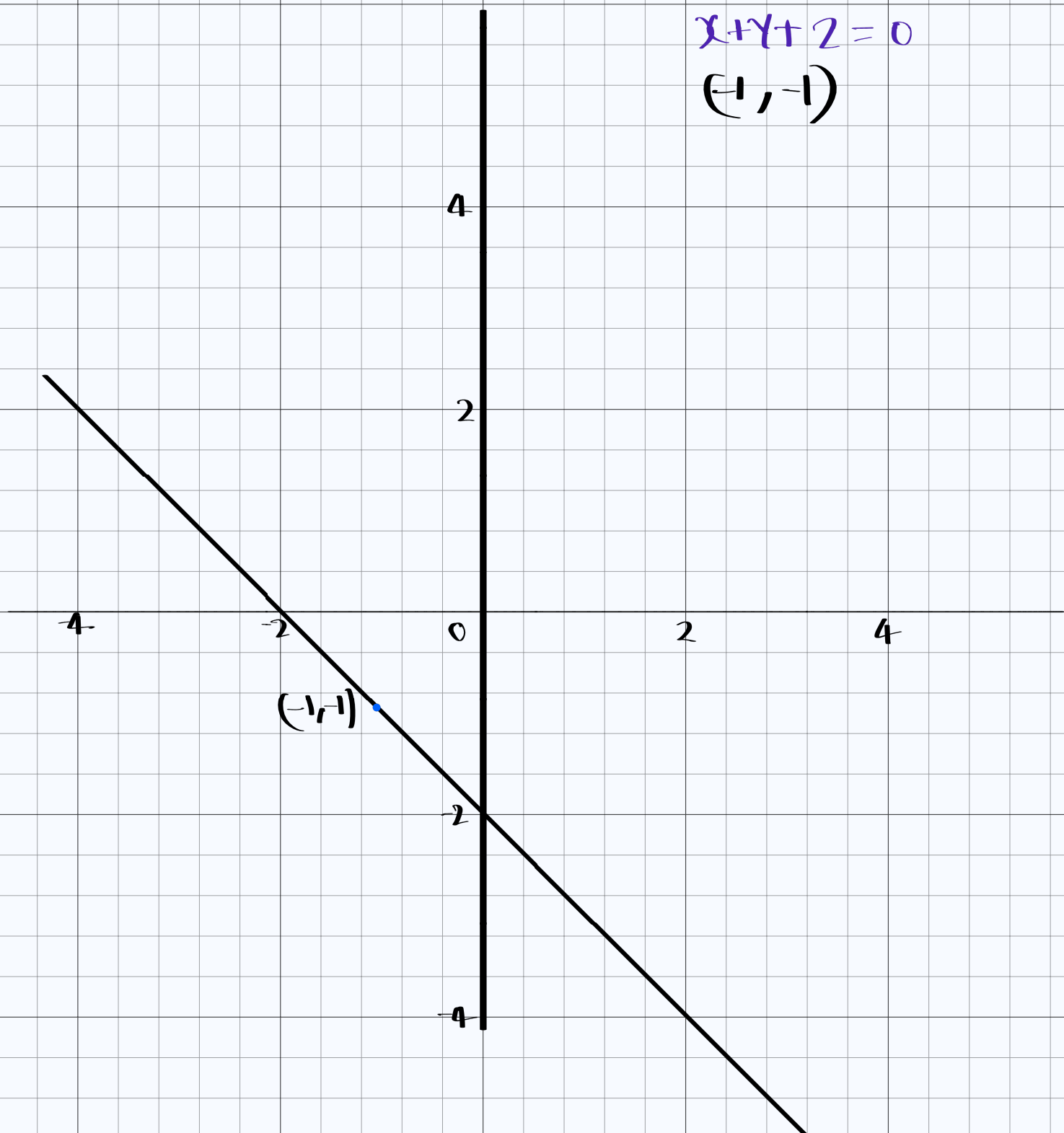
$$= \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow \hat{i} + \hat{j}$$

→ If a point x' lies on line
 $w^T x + w_0 = 0$ then it
 satisfy

$$w^T x' + w_0 = 0$$

N.P. → If it is not lying on
 line that means either it
 is greater than zero or smaller

an zero.



→ Consider 2 Vectors $x = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ and $y = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$

What is the dot product?
What is the angle b/w them?

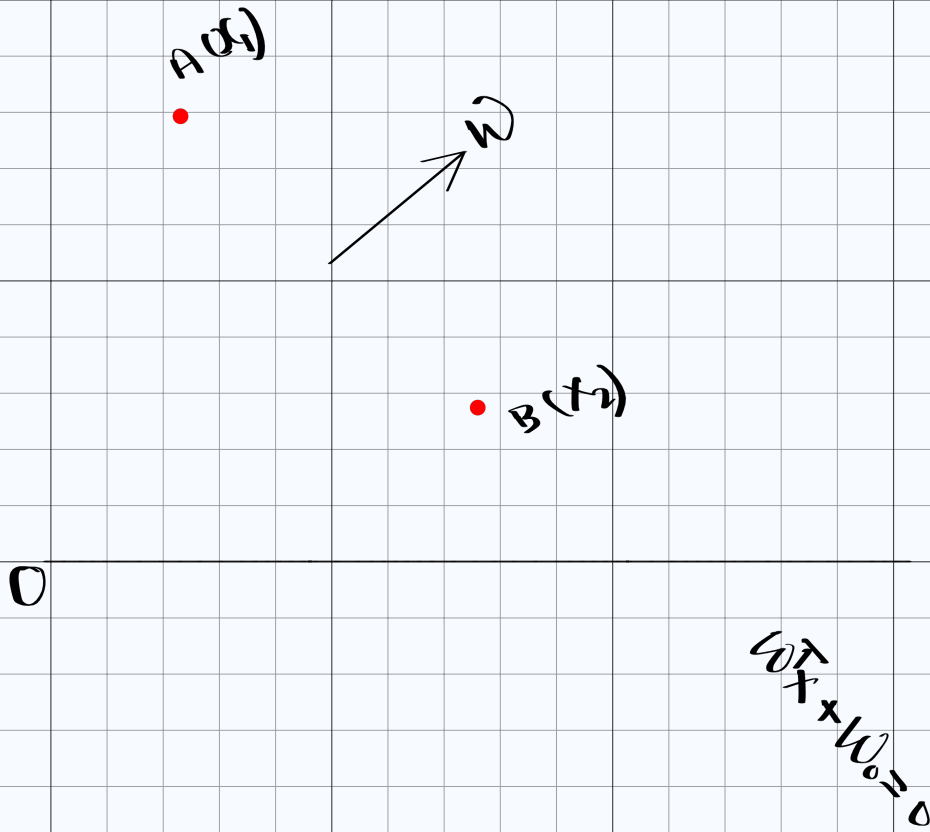
Solⁿ dot product = $2 \times 3 + (-2) \times 3$
 $6 - 6$

$$\text{angle} = \cos^{-1}(0) = 90^\circ$$

Above everything was taught to

→ Perceive that $\vec{w} \perp$ to line/
hyperplane

$$w^T x + w_0 = 0$$



$$w^T x_1 + w_0 = 0 \rightarrow \textcircled{I}$$

$$w^T x_2 + w_0 = 0 \rightarrow \textcircled{II}$$

Subtract $\textcircled{I}$ from $\textcircled{II}$

$$w^T (x_2 - x_1) = 0$$

$$\boxed{w^T \vec{AB} = 0}$$

$$\vec{w} \perp \vec{AB}$$

$$\vec{w} \perp w^T x + w_0 = 0$$